



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014

ANSWERS

BIOLOGY

Higher 2

Paper 2 Core Paper

9648/02

Thursday, 28 August

2 hours

Section A [Part 1]

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an electron micrograph of part of a liver cell.

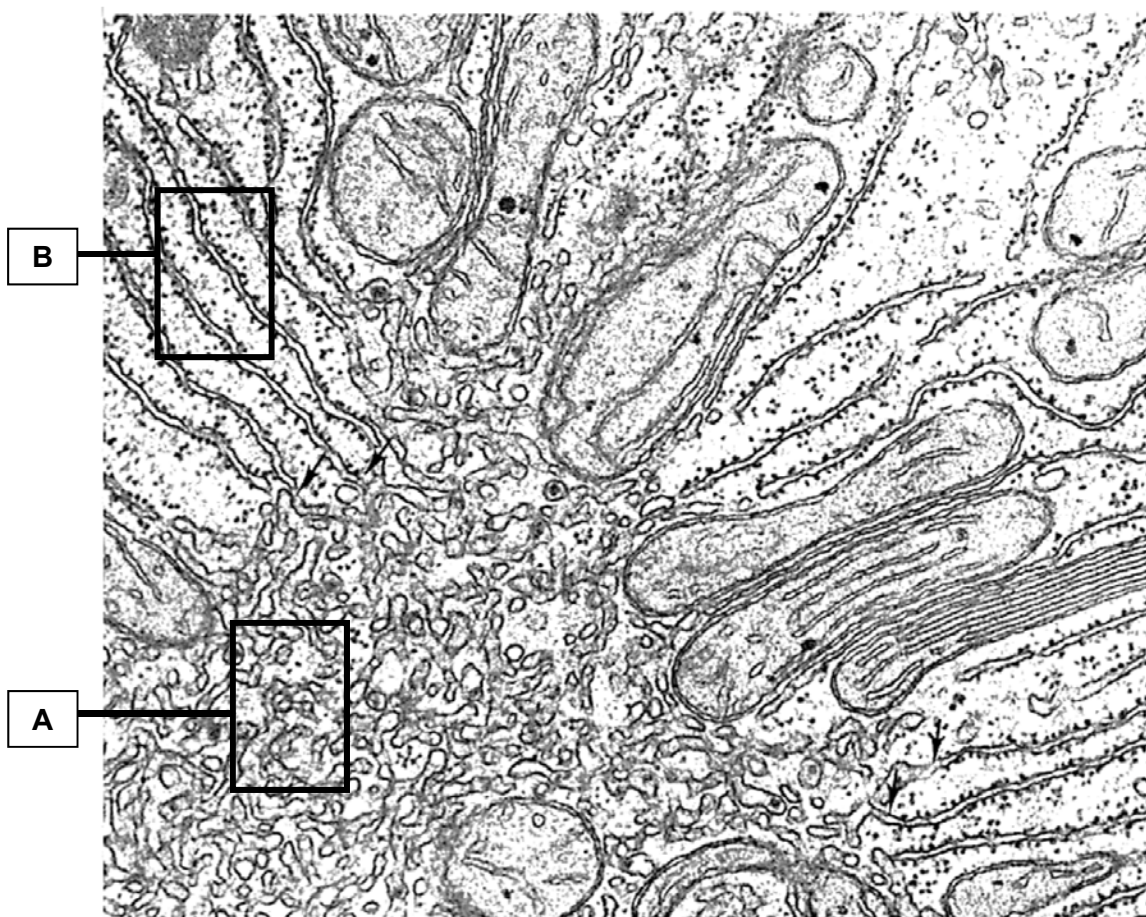


Fig. 1.1

- (a) Identify structures **A** and **B**. [2]

A: [smooth endoplasmic reticulum](#)

B: [rough endoplasmic reticulum](#)

(b) State the function of structure A. [1]

Concerned primarily with the synthesis of lipids (including membrane phospholipids), cholesterol and steroid hormones.

(c) Relate the structural features of B to its function. [2]

1a) Structure: Ribosomes are present on its surface.

1b) Function: This is the site where proteins are made and transported through the cisternae of the ER.

2a) Structure: B consists of a network of membranous tubules and sacs / called cisternae.

2b) Function: Transportation of the protein product takes place: through the cisternae to its destination, then packaged in membranous vesicles and exported / dispatched to other parts of the cell

Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 1.2 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

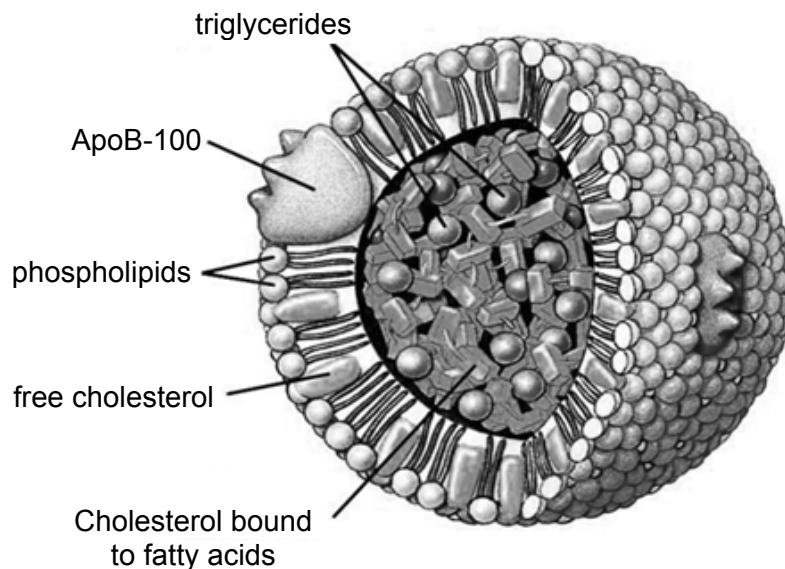


Fig. 1.2

(d) With reference to Fig. 1.2, outline the roles of lipids in the LDL coat. [3]

Cholesterol and phospholipids are important components of the LDL coat:

1. Free cholesterol – maintains fluidity of the LDL coat.
2. It disturbs the close packing of phospholipids thus keeping them more fluid.
3. Phospholipids – major constituent of the LDL coat, they form the monolayer in LDL coat.

4. The role of the phospholipid monolayer is to form a boundary between the hydrophobic triglyceride core from the aqueous exterior of the LDL.
5. The hydrophobic tails face inwards and make contact with the hydrophobic triglyceride core
6. The hydrophilic phosphate heads face outwards and make contact with the exterior aqueous environment.

(e) Fig. 1.3 shows the three dimensional structure of ApoB-100.

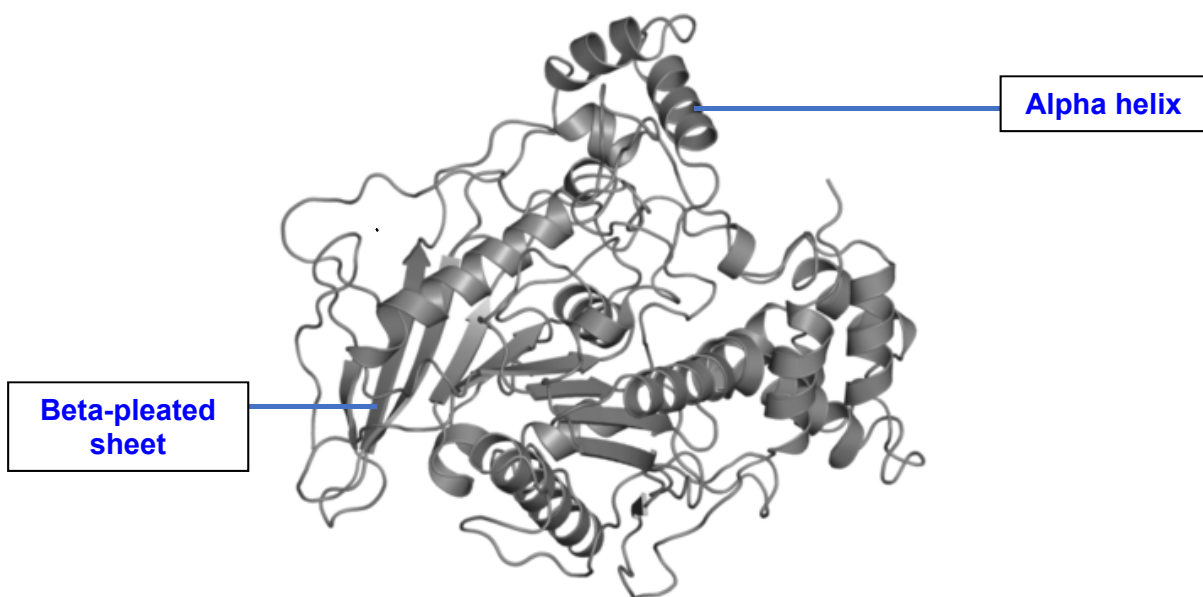


Fig. 1.3

- (i) Label the type(s) of secondary protein structure observed in Fig. 1.3. [1]
 - (ii) Explain how the structure of ApoB-100 protein is maintained. [3]
1. The primary sequence of the ApoB-100 protein is held by peptide bonds between amino acid residues;
 2. The secondary structure consisting of the α -helices and β -pleated sheets
 3. are maintained by hydrogen bonds between peptide linkages of the polypeptide backbone – the R groups of amino acids are not involved.
 4. The tertiary structure of the ApoB-100 protein is maintained by the interactions among the R groups of amino acids at different regions in the polypeptide chain –
 5. namely the disulphide bond, the hydrogen bond, the ionic bond and hydrophobic interactions.
 6. ApoB-100 folds into a compact, globular 3-dimensional structure.

[Total: 12]

- 2 Fig. 2.1 shows the mode of action of the sodium-potassium **pump** found on the cell surface membranes of sweat glands.

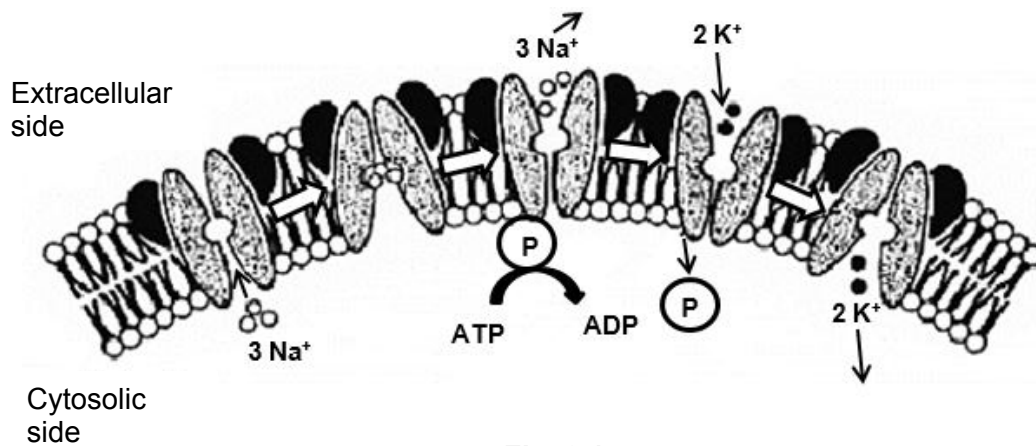


Fig. 2.1

(a) Explain why the cell surface membrane is referred to as the 'fluid mosaic model'. [2]

1. Protein molecules float about in a fluid phospholipid bilayer,
2. with only their hydrophilic portions exposed to water, in constant motion. (idea of free movement)
3. The scattered protein molecules resemble a mosaic, but since the phospholipid bilayer is fluid, the proteins form a fluid mosaic pattern.

(b) Explain why ions can only move across cell surface membranes via transmembrane proteins. [2]

- 1 Ions are charged particles
- 2 and are unable to pass through the hydrophobic interior of the cell membrane
- 3 due to the fatty acid tails.
- 4 The transmembrane proteins provide a hydrophilic channel for ions to pass through.

OR

The interior of the transmembrane protein is lined with amino acid hydrophilic R groups

The sodium-potassium pump, also known as sodium-potassium **ATPase**, is able to catalyse the hydrolysis of ATP molecules to ADP and P_i . This reaction is coupled to the movement of ions across the membranes.

Movement of ions is based on the initial binding of Na^+ ions to the pump. This stimulates phosphorylation of the pump which causes a change in its conformation, leading to ion movement.

(c) With reference to Fig. 2.1,

- (i) identify the type of transport mechanism that enables sodium ions to leave the cell; [1]

Active Transport

- (ii) explain how the pump is also able to catalyse the hydrolysis of ATP molecules leading to the movement of ions across the membranes.

- 1 ATPase acts as the enzyme which
- 2 has an active site whose shape is complementary to the shape of substrate / ATP.
- 3 ATP will bind to the active site and the catalytic residues at the active site will act on the bonds in ATP to form ADP and P_i .
- 4 The energy released from the hydrolysis of ATP
- 5 will cause a conformational change to the (transmembrane) protein
- 6 Allowing 3 Na^+ ions to leave the cell and 2 K^+ ions to enter the cell.

Ouabain is a cardiac glycoside that is able to affect the sodium-potassium pump. In an experiment to determine the effect of ouabain on the pump, sweat glands from rhesus monkeys were taken and treated to measure ATP hydrolysis.

For each of the treatments, 5 sweat gland units were placed in a mixture of buffer and ATP.

Treatment 1 : Sweat glands of rhesus monkey with Na^+ and K^+ added.

Treatment 2 : Sweat glands of rhesus monkey with Na^+ , K^+ and ouabain added.

Treatment 3 : Sweat glands of rhesus monkey without additional Na^+ and K^+ .

Table 2.1 shows the data collected at the end of the 1 hour experiment.

Table 2.1

Treatment	Number of moles of ATP hydrolysed g^{-1}
1	0.58
2	0.37
3	0.38

(d) Explain the effects of ouabain on ATP hydrolysis.

1. Ouabain is an inhibitor of the enzyme sodium-potassium ATPase
2. QF : Comparing treatments 1 and 2, there is a decrease in the number of moles of ATP hydrolysed from 0.58 moles/g for the treatment with Na⁺ and K⁺ only as compared to 0.37 moles/g for the treatment with Na⁺ and K⁺ with ouabain added.

(e) Suggest a reason for the results of Treatment 3.

1. Lower than Treatment 1, at 0.38 moles / g as there was absence of Na⁺ and K⁺.
2. The existing Na⁺ within the cell will be depleted and no additional Na⁺ will be transported in via facilitated diffusion. Reduced phosphorylation hence hydrolysis of ATP is reduced.

[Total: 11]

- 3 Eggs and early embryos of amphibians are particularly suitable for biochemical studies of the cell cycle. These organisms typically have large eggs and fertilization is followed by multiple synchronous cell cycles.

As oocytes develop in the frog ovary, they become arrested in G₂ for 8 months, during which time they grow in size to a diameter of 1 mm. When stimulated by a male frog, the female's ovarian cells will secrete progesterone, which induces the oocytes to undergo meiosis I.

The oocytes will progress to Metaphase II, where they are arrested and await fertilization. Meiosis is completed only when the oocyte is fertilized by a sperm. Subsequently the first mitotic division of embryogenesis occurs. The entire process of oocyte maturation is shown in Fig. 3.1.

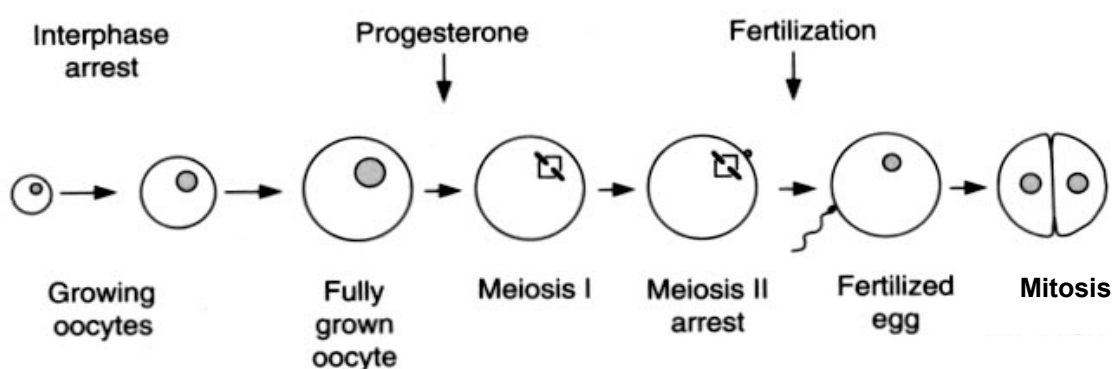


Fig. 3.1

(a) G₂ is part of a stage that takes place during the cell cycle. Describe what happens during this stage. [3]

1. Interphase
2. G₁/ first “gap” phase – The cell grows and builds up a large store of materials for mitosis by making proteins for replication, biochemical and new organelles
3. S/Synthesis phase – Replication of DNA occurs. This doubles the DNA content of the cell
4. G₂ / second “gap” phase; Synthesis of proteins and Formation of new organelles

(b) Distinguish the significance of meiosis from mitosis. [3]

Meiosis	Mitosis
1. brings about <u>genetic variation</u>	<u>maintains the genetic stability of an organism/ or a cell from one generation to the next</u>
2. Daughter cells are <u>genetically different</u>	2 daughter cells formed: ➤ are <u>genetically identical</u> to the parent cell as ➤ they contain the <u>same number and types of chromosomes</u> as the original parent nucleus.
3. Essential to the process of <u>evolution</u> . By providing a varied stock of individuals, it allows the <u>natural selection</u> of individuals best suited to survive under existing environmental conditions. This ensures that the species would be <u>able to adapt even when environmental conditions change</u> .	➤ Does not contribute to the process of evolution.

(c) Suggest why eggs of amphibians are suitable for studies of the cell cycle. [2]

1. Large eggs – easier to manipulate /observe the processes/OWTTE
2. Large numbers of eggs can be prepared that are all proceeding synchronously using methods of progesterone treatment / fertilization

The Cama was created in a laboratory in Dubai, from breeding a male camel with a lama, with the purpose of higher wool production and easier temperament. Interestingly the cama, unlike most other types of animal hybrids, are always fertile.

(d) Suggest why most animal hybrids are sterile. [3]

1. Unable to form functional gametes due to/ no gametes formed
2. Different chromosome number between the parental species

OR

Different chromosomal constitution – different genes

3. Bivalents cannot form/ No pairing up of homologous chromosomes during Prophase I
4. Thus unequal separation of chromosomes/ chromatids [1/2] during Anaphase I/ Anaphase II

[Total: 11]

- 4 In bacteria, genes may be organised into operons. The *trp* operon consists of genes that encode five enzymes involved in the synthesis of tryptophan (*trp*). The *trp* operon and its regulatory gene are shown in Fig. 4.1.

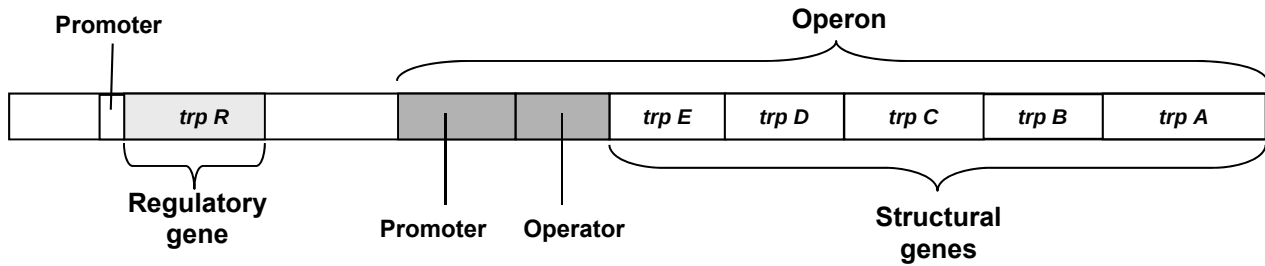


Fig. 4.1

(a) With reference to the *trp* operon, explain what is meant by a *repressible operon*. [3]

1. A repressible operon is one where its transcription is usually turned on
2. but can be inhibited / repressed when a specific small molecule / tryptophan binds allosterically to its regulatory protein.
3. When tryptophan is present, it acts as a corepressor, binds to repressor.
4. Change shape of inactive repressor protein to the active conformation.
5. The active repressor now binds to the operator.
6. This blocks the binding of RNA polymerase to promoter,
7. The *trp* operon genes are not transcribed. Enzymes for tryptophan synthesis are not produced.

Different operons in the bacteria encode different types of enzymes.

(b) State **one** difference between a repressible enzyme and an inducible enzyme. [1]

Any one:

- Inducible enzymes are encoded by inducible operons while repressible enzymes are encoded by repressible operons
- Inducible enzymes usually function in catabolic pathways which break a nutrient down to simpler molecules while repressible enzymes usually function in anabolic pathways which synthesize essential end products from raw materials (precursors).
- The synthesis of inducible enzymes are induced by the presence of their substrate (eg. Inducer is allolactose) while the synthesis of repressible enzymes is repressed by the presence of the end product (tryptophan) of the pathway.
- Inducible enzymes are usually absent when inducer is absent while repressible enzymes are always present unless the end product (tryptophan) of their pathway is available.

Table 4.1 indicates the activity levels of the functional enzymes **E**, **D**, **C**, **B** and **A** in the presence and absence of tryptophan (Trp) in the culture medium of the different types of bacteria cells. The mutant strain is the result of a single base-pair substitution in a single component of the *trp* operon.

Table 4.1

Enzymes	Activity level of enzymes / arbitrary units							
	Wild-type		Mutant strain 1		Mutant strain 2		Mutant strain 3	
	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present
E	700	0	700	700	700	0	0	0
D	700	0	700	700	0	0	0	0
C	700	0	700	700	700	0	0	0
B	700	0	700	700	700	0	0	0
A	700	0	700	700	700	0	0	0

(c) Identify and explain which mutant strain has a phenotype that is consistent with a loss-of-function mutation in the *trp R* gene. [2]

1. Mutant strain 1
2. Mutant strain 1 shows activity level of enzymes is 700a.u. whether tryptophan is present or absent / when tryptophan is present.
3. This indicates that the repressor was non-functional / not produced due to loss of function mutation
4. Even if tryptophan is present, there is binding of tryptophan to repressor does not take place so binding of repressor to operator does not take place

(d) Identify where the loss-of-function mutation occurred in the *trp* operon of mutant strain 3. [1]

- **Promoter**

(e) The eukaryote genome differs from the prokaryote genome in that some genes may be present in multiple copies as a result of gene amplification.

Describe the significance of gene amplification during development. [2]

1. Gene amplification is an increase in gene expression / allows more copies of mRNA to be made.

- Reject suitable amounts or regulate amounts – too vague
- No marks for stating extra copies of the gene – because it is already given in the question.

2. This leads to an increased amount of gene product in a short period of time; (idea of increased rate)
3. More polypeptide synthesized to carry out specialised functions in the cell / for construction of certain structures of the cell / provide a suitable e.g.
4. The effect is only temporary.

- (f) State **one** difference between the control of gene expression in eukaryotes and prokaryotes at the transcriptional level. [1]

Any one:

FEATURE	PROKARYOTIC GENOME	EUKARYOTIC GENOME
Chromatin Modification	Does not occur: DNA demethylation is for protection purposes only. E.g. prevention of DNA degradation when viruses invade	DNA demethylation: for easier access of transcription factors to genes Histone acetylation: for easier access of transcription factors to genes
Global Regulatory Systems	Common protein regulating a collection of genes that may be scattered across bacterial chromosome	Does not occur
Strength of Promoters	The stronger the promoter, the higher the frequency of transcription initiation at each promoter, the higher the level of transcription	Does not occur
Access to Regulatory Sequences/ Control Elements	Operator: situated between promoter and first structural gene Activator binding site: situated before promoter	Proximal control elements: situated near transcription unit Distal control elements: situated thousands of nucleotides away from gene and upstream or downstream from transcription unit.
Availability of Proteins that response to External Stimuli	Regulatory proteins (activators or repressors) binds to regulatory sequences	Specific transcription factors (activators) binds to Distal control elements
Different types of Sigma Factors	When a complex process requires the synthesis of several gene products in a precisely timed sequence, transcription may be regulated by a series of sigma factors, which recognize specific promoters	Does not involve sigma factors.
Attenuation	- Premature transcription termination depends on how the translational machinery reads the secondary structure of the mRNA leader - Only possible because transcription and translation are coupled in prokaryotes	Attenuation does not occur

[Total: 10]

- 5 Fig. 5.1 shows a transmission electron micrograph of a bacterium.



(source: http://cronodon.com/BioTech/Bacteria_Growth.html)

Fig. 5.1

- (a) Describe and explain the significance of the process occurring in Fig. 5.1. [3]

1. **Binary fission**
2. **The cell is partitioned by the inward growth of new cell membrane**
3. **and cell wall material approximately midpoint of the cell called a septum**
4. **Asexual reproduction**
5. **offspring are genetically identical to the parent**
6. **Rapid increase in cell numbers**

This particular strain of bacteria carries an extra piece of genetic information encoded on an F plasmid.

- (b) Compare the F plasmid with the bacterial chromosome. [3]

Similarities

1. Both are made up of **circular, double stranded DNA molecules**
2. Both have only **one origin of replication**

Differences

1. Plasmids contain **less genes/smaller in size**, while the bacterial chromosome contains **more genes/bigger in size**.
2. **Specifically for F plasmid: Genes on the F plasmid direct the synthesis of sex pili** (singular: pilus) for **conjugation** to take place whereas the bacterial chromosomes do not contain the genes which code for the formation of the sex pili

It has been shown that there is an increase in prevalence of antibiotic resistance among *E. coli* strains in Asia, and majority of the antibiotic resistances were due to the acquisition of antibiotic-resistance genes.

The effects of antibiotic **B** were studied on two strains of *E. coli*. The bacteria were cultured with antibiotic **B** and without antibiotic **B**. Fig. 5.2 shows the change in the number of bacteria cells.

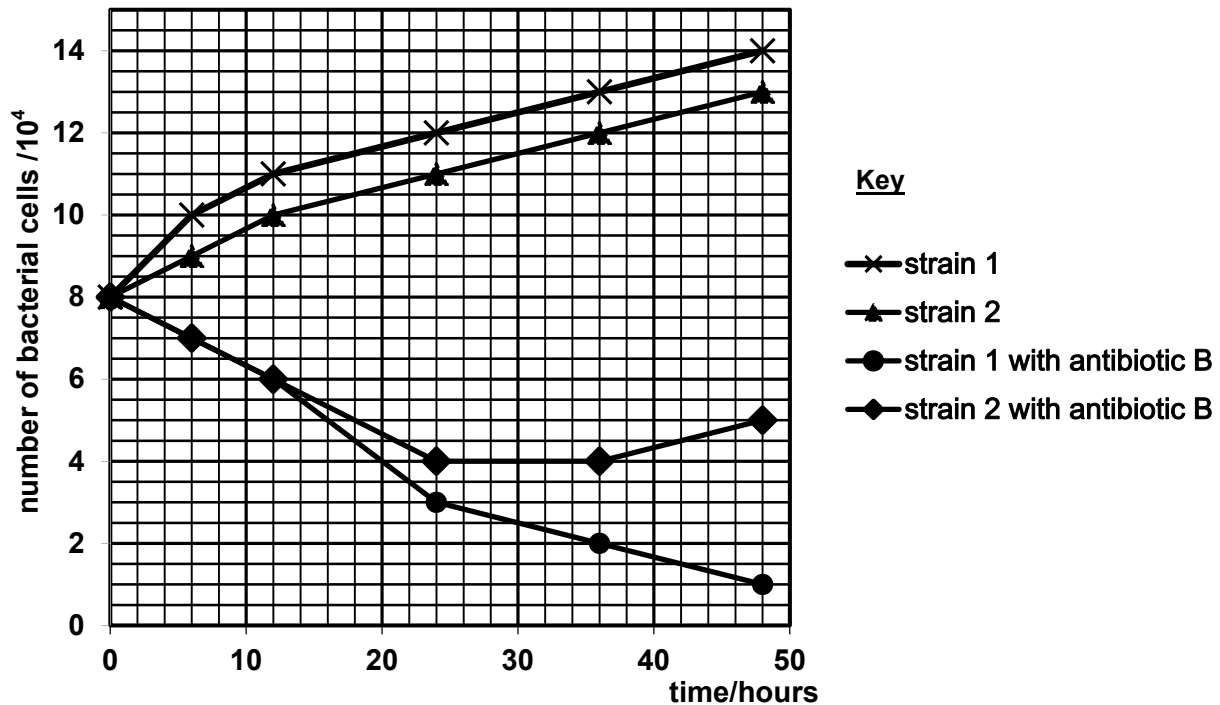


Fig. 5.2

(c) With reference to Fig. 5.2,

(i) describe the effect of antibiotic **B** on strain 1; [2]

- 1a. **G.S. : Antibiotic treatment** causes a decrease in the number of bacterial cells for strain 1
- 1b. **Q.F. : As time increased from 0 to 48 hours, the number of bacterial cells in strain 1 decreased** from 8×10^4 to 1×10^4 in the presence of antibiotic B
- 1c. In the absence of antibiotic B the number of bacterial cells increased from 8×10^4 to 14×10^4

(ii) explain which strain contains the antibiotic resistance gene. [2]

1. strain 2 contains the antibiotic resistance gene
2. As time increased from 36 to 48 hours, the number of bacterial cells for strain 1 decreased in the presence of antibiotic B from 2×10^4 to 1×10^4
3. while the number of bacterial cells for strain 2 increased in the presence of antibiotic B from 4×10^4 to 5×10^4

4. the increase in the number of bacterial cells is due to binary fission/division/reproduction of the cells which contain the gene for antibiotic resistance, OWTTE

[Total: 10]

- 6 In the inheritance of feather colour in parakeets, there is an interaction between two autosomal gene loci, **A/a** and **B/b**. Parakeets carrying only the dominant allele, **A**, have yellow plumage while parakeets carrying only the dominant allele **B** have blue plumage. Parakeets with at least one dominant allele at both gene loci appear to have green plumage.

A parakeet with green plumage was crossed with a white parakeet. All the resulting progeny have green plumage. When pairs of the offspring were allowed to interbreed, they produced the following results in Table 6.1.

Table 6.1

Plumage colour	Number
Green	37
Yellow	11
Blue	12
White	4

- (a) Identify the type of gene interaction. [1]

Gene Collaboration

- (b) State the possible genotypes of parakeets with blue plumage. [1]

aaBB, aaBb

(c) Draw a genetic diagram to explain the results of the cross **between the offspring**. [4]

Key:

A represents the dominant allele that codes for yellow plumage

B represents the dominant allele that codes for blue plumage

F1 phenotypes : green plumage x green plumage

F1 genotype: AaBb x AaBb

Gametes :



Punnett square:

	AB	Ab	aB	ab
AB	AABB Green plumage	AABb Green plumage	AaBB Green plumage	AaBb Green plumage
Ab	AABb Green plumage	AAbb Yellow plumage	AaBb Green plumage	Aabb Yellow plumage
aB	AaBB Green plumage	AaBb Green plumage	aaBB blue plumage	aaBb blue plumage
ab	AaBb Green plumage	Aabb Yellow plumage	aaBb Blue plumage	aabb White plumage

F2 phenotypes : green plumage : yellow plumage : blue plumage : white plumage

F2 phenotypic ratio: 9 : 3 : 3 : 1

Observed nos: 37 : 11 : 12 : 4

- (d) Calculate the chi-squared (χ^2) value on the data to determine the probability of the results of the cross between the offspring departing significantly by chance from the expected ratio. [2]

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

	Expected numbers	Observed numbers	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Green plumage	36	37	1	1	0.03
Yellow plumage	12	11	-1	1	0.08
Blue plumage	12	12	0	0	0
White plumage	4	4	0	0	0

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= 0.03 + 0.08 + 0 + 0$$

$$= 0.11 \text{ (to 2 dp)}$$

OR

$$= \frac{(37 - 36)^2}{36} + \frac{(11 - 12)^2}{12} + \frac{(12 - 12)^2}{12} + \frac{(4 - 4)^2}{4}$$

$$= 0.12$$

- (e) Explain if the results of the cross conform to the expected ratio. [2]

1. For 3 degrees of freedom,
2. the χ^2 value of 0.11 is less than 7.82,
3. therefore the probability that the deviation is due to chance is > 0.05,
4. allowing us to conclude that the deviation is statistically insignificant and is due to chance alone. Therefore, the results of the cross conform to the expected ratio.

[Total: 10]

- 7 There are many species of the Anole lizards throughout the Caribbean and the surrounding mainland. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida.

Fig. 7.1 shows the distribution of four species of anole lizards.

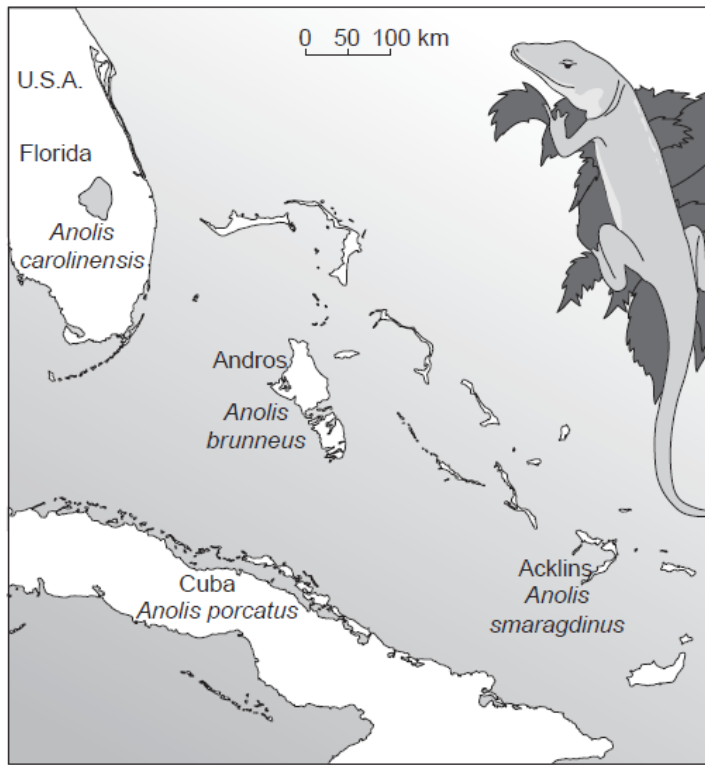


Fig. 7.1

An investigation using DNA analysis was carried out to establish the relationships between the four lizard species, using DNA analysis. The base sequences of a region of mitochondrial DNA from the four species were compared. The percentage differences in base sequences are shown in Table 7.1.

Table 7.1

	<i>A. brunneus</i>			
<i>A. brunneus</i>		<i>A. smaragdinus</i>		
<i>A. smaragdinus</i>	12.1		<i>A. carolinensis</i>	
<i>A. carolinensis</i>	16.7	15.0		<i>A. porcatus</i>
<i>A. porcatus</i>	11.3	8.9	13.2	

(a) With reference to Table 7.1,

(i) state the species to which *A. brunneus* appears to be least closely related; [1]

- *Anolis carolinensis*

[accept as stated in the table *A. carolinensis*]

(ii) explain your answer in (a) (i). [1]

- *Anolis carolinensis* has the largest percentage of nucleotide differences of 16.7 when compared to *Anolis brunneus*.

(b) The researchers put forth the hypothesis that *A. brunneus*, *A. smaragdinus* and *A. carolinensis* all originated from *A. porcatus* in three separate events.

(i) Explain how the results in Table 7.1 support the hypothesis; [3]

1. Each species is more closely related to *A. porcatus* than to each other;
2. G.S : *C carolinensis* have smaller nucleotide differences with *A. porcatus* than with the other species;
3. QF: *A. brunneus* only has 11.3 differences with *A. porcatus* but 16.7 differences with *A carolinensis* and 12.1 differences with *A. smaragdinus*
4. QF : *A. smaragdinus* has only 8.9 differences with *A. porcatus* but 15.0 differences with *A. carolinensis*.
5. QF : *A. carolinensis* has only 13.2 differences with *A. porcatus*.
6. The 1st separation event resulted in *A. carolinensis*, 2nd separation resulted in to *A. brunneus* and 3rd separation event resulted in *A. smaragdinus*.

(ii) Explain how a population of *A. porcatus* that became isolated on an island could evolve into each of the new species. [4]

1. Allopatric speciation took place ;
2. Due to geographical/physical barrier between different lizard populations
3. The lizard populations were separated by water.
4. There was no breeding / gene flow between populations.
5. Isolated populations undergo genotypic and / or phenotypic changes due to spontaneous mutation / Mutations occurred in each population gave rise to new alleles
6. Different selection pressures / different (environmental) conditions in each island;
7. Different alleles selected for with an increase in frequency of favourable alleles.

8. Genetic drift / loss of alleles from population gene pool due to chance event.
9. Over time each population changed so much that they are reproductively isolated and are no longer capable of exchanging genes even if contact is restored.

[Total: 9]

8 Fig. 8.1 shows an electron micrograph of a cholinergic synapse.

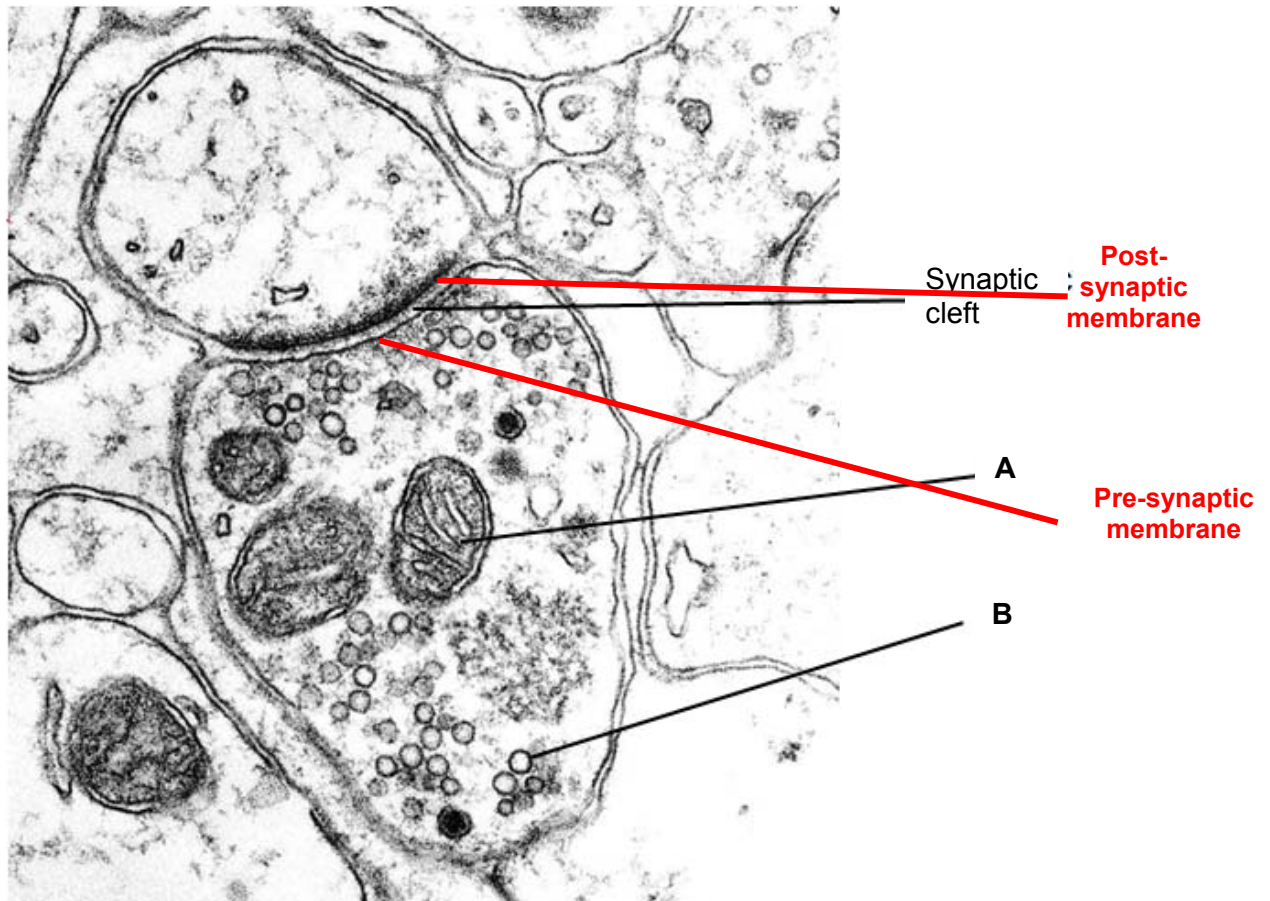


Fig. 8.1

- (a) Label the pre-synaptic membrane and post-synaptic membrane on Fig. 8.1. [1]
- (b) Identify what is present in structure B.

Acetylcholine

(c) Explain the role for structure **A** in the transmission of a nerve impulse across the synapse.

1. **Structure A which is the mitochondria**
2. **which synthesizes ATP.**
3. **ATP is also needed in the movement of synaptic vesicles to release the acetylcholine into the synaptic cleft (OWTTE)**
4. **by exocytosis. (MUST HAVE!)**
5. **– calcium pump removing Ca^{2+} from the presynaptic knob.**
6. **– reabsorption of neurotransmitters(choline) back into the pre- synaptic knob**

Botulinum toxin is a neurotoxin that interferes with neural transmission, causing muscle paralysis. Botulinum can be used clinically by botox users to reduce wrinkles.

The toxin is a protease which cleaves proteins called SNAREs. The primary role of the SNARE protein is to mediate fusion of vesicles to membranes.

(d) Explain how botulinum affects transmission of nerve impulse across the synapse.

1. **When the Botulinum toxin cleaves the SNARE protein, it is no longer functional and is not able to mediate fusion of the synaptic vesicles to the membranes (OWTTE)**
2. **This will prevent the release of acetylcholine from the synaptic vesicle into the synaptic cleft.**
3. **No acetylcholine will bind to receptors on the post-synaptic membrane,**
4. **and the ligand-gated sodium channels will not open. (preventing the transfer of chemical signal to the muscle / motor neurone)**

(e) Suggest how botox is able to reduce wrinkles.

1. **Botulinum paralyses the facial muscles that cause the wrinkle in the skin.**

[Total: 7]

END OF SECTION A

Section B: Essay Questions

- 9 (a) Describe how cellulose differs from starch. [5]

Structure:

1. Cellulose is made up of β glucose units while starch is made up of α glucose units
2. $\beta(1,4)$ glycosidic bond between glucose units in cellulose while $\alpha(1,4)$ glycosidic bond between glucose units in starch
3. In cellulose, alternate glucose units in a chain are rotated at 180° but glucose units are not rotated in the glucose chains in starch.
4. In cellulose, the chains remain straight, while in starch, each chain form a helix.
5. In cellulose, hydroxyl groups of each chain project outwards (thereby allowing cross-linking between chains) forming microfibril and macrofibril while in starch, hydroxyl groups projects inside the helix. (thereby no cross-linking between chains)

Function:

6. Cellulose has high tensile strength and functions as a structural material for plant cell wall. Starch being compact with a large number of glucose units functions well as a storage carbohydrate in plant cells.

- (b) Describe how carbohydrates are formed in the Calvin cycle. [6]

1. CO_2 combines with ribulose biphosphate (RuBP).
2. The enzyme involved is RuBP carboxylase
3. The result of this combination is an unstable 6 carbon compound,
4. It spontaneously breaks down to give two molecules of glycerate-3-phosphate (GP)
5. GP is reduced to glyceraldehyde-3-phosphate (GALP/G3P).
6. The H^+ for the reduction comes from NADPH.
7. The energy for this step comes from ATP
8. Two GALP molecules combine to form a 6C sugar/glucose
9. It can be converted to starch for storage.
10. Most of the GALP enters a series of reactions, driven by ATP, which results in the regeneration of RuBP. and this allows CO_2 fixation to continue.
11. 3 CO_2 molecules, 9 ATP molecules and 6 NADPH molecules are required to form one 3-carbon GALP per cycle
12. It will combine with another GALP from another turn of the Calvin cycle to produce a 6C sugar, which will later be converted into glucose.

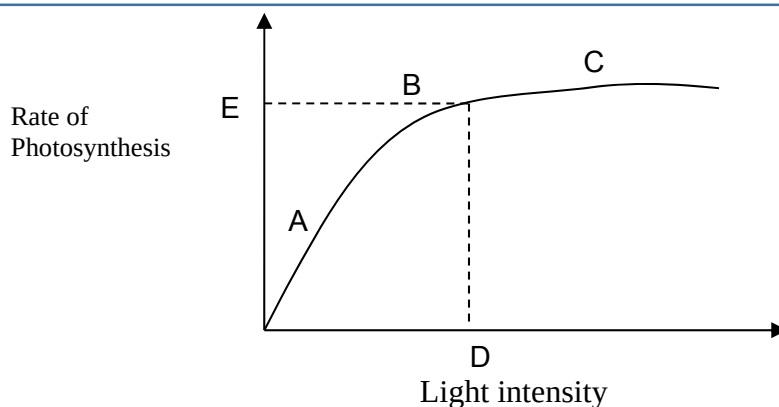
- (c) Explain the effect of light and carbon dioxide as limiting factors on the rate of photosynthesis. [9]

1. When a biochemical process is affected by several factors, its rate is limited by that factor which is nearest its minimum value and the factor that directly affects the process if its quantity is changed, and is therefore the limiting factor

Light intensity

2. At low light intensities, rate of photosynthesis increases (linearly) with increasing light intensity.
3. The rate at which ATP and NADPH are produced is too slow to allow the dark reactions to proceed at maximum rate.
4. Hence, light is a limiting factor in photosynthesis at low light intensities.

Referring to the figure,



At region A, light intensity is the limiting factor (same as point 4)

5. At region B, another factor (eg temperature or CO₂ concentration) other than light intensity is becoming the limiting factor
6. At region C, light intensity is no longer the limiting factor – light saturation occurs at C. The light saturation point occurs at D – beyond this point, a further increase in light intensity does not result in an increase in the rate of photosynthesis

In the natural environment, on a clear summer day, illumination is about 100 000 lux, whereas the light saturation for C3 plants is reached at about 10 000 lux – about 10%. Light is rarely a limiting factor during the day.

Hence, in the natural environment, except in early morning and early evening, and for shaded plants, light is normally not a limiting factor in photosynthesis.

Compensation point

7. This is the point at which the rate of photosynthesis is equal to the rate of respiration.
8. At this point, all the CO₂ produced during respiration is used for photosynthesis, and all the O₂ produced during photosynthesis is used for respiration. Therefore, there will not be any gaseous exchange between the plant and its environment.

9. The compensation point is reached at rather low light intensity, usually at sunrise and sunset.
10. Light intensity is also a direct factor for stomata opening. Stomata opens as light intensity increases early each day, before light levels are saturating.

Light quality/wavelength

11. The effect of light quality is revealed by the action spectrum for photosynthesis (i.e. which wavelengths are utilized for photosynthesis) and occurs at the highest rate at 680-700nm and 400-450nm wavelength of light.

Carbon dioxide concentration

12. CO₂ is needed for the dark reactions.
13. Under normal field conditions, CO₂ is the major limiting factor in photosynthesis, since its concentration in the atmosphere is about 0.03 – 0.04 %.
14. Increasing CO₂ concentration up to 0.5% leads to a linear increase in photosynthetic rate BUT injures the plant.
15. As carbon dioxide concentrations rise, the rate at which sugars are made by the light-independent reactions increases until limited by other factors.
16. Rubisco, the enzyme that captures carbon dioxide in the light-independent reactions, has a binding affinity for both carbon dioxide and oxygen.
17. When the concentration of carbon dioxide is high, Rubisco will fix carbon dioxide.
18. If the carbon dioxide concentration is low, Rubisco will bind oxygen instead of carbon dioxide. (This process, called photorespiration, uses energy, but does not produce sugars.)

10 (a) Describe how the structure of transfer RNA is adapted to its function. [5]

1. tRNA plays an important role in amino acid activation.
2. The attachment of amino acid to its specific tRNA is mediated by a group of enzymes known as aminoacyl tRNA synthetases.

Structural features	Significance and contribution to function of molecule
3. CCA site at 3' end	▪ For attachment of <u>activated amino acid</u> to form an amino acyl-tRNA complex. ▪ Allows tRNA molecule to carry its specific corresponding <u>amino acid to the ribosome during protein synthesis</u>.
4. Anticodon of specific triplet base sequence (3 bases)	▪ Allows <u>complementary base pairing between anti-codon of tRNA with codon of mRNA</u> ▪ This ensures the correct amino acyl-tRNA complexes occupy the 'P' and 'A' site of the ribosome for protein synthesis. ▪ This allows for correct sequencing of amino acids on the polypeptide chain coded for by the gene.
5. <u>Clover-leaf shape, single stranded with 3 loops & a stem held together by hydrogen bonds</u>	• Has shape complementary to active site of aminoacyl-tRNA synthetase for amino acid activation

- (b) Describe how the information on an mRNA is used to synthesize a polypeptide chain.
[8]

Initiation of Translation – Binding of 1st amino acyl-tRNA complex to mRNA \

1. An amino acyl-tRNA complex with the anticodon UAC binds to the start codon (AUG) on mRNA. This complex is positioned at the “P” site of the ribosome.
2. tRNA with the anticodon UAC always carries the amino acid methionine.
3. Methionine is always the first amino acid in a polypeptide chain.
4. Now the large subunit of the ribosome binds to the small subunit.

Chain elongation – Binding of second amino acyl-tRNA complex to Mrna

5. The second amino acyl-tRNA complex binds to mRNA. It has the anticodon complementary to the second codon on mRNA.
6. The complex is held at the “A” site of the ribosome.

Chain elongation – Transfer

7. Formation of a peptide bond between the first and the second amino acids then occurs, utilising energy from hydrolysis of ATP.
8. This is catalysed by the peptidyl-transferase which is present on the large subunit of the ribosome.

Chain elongation – Translocation

9. Once the peptide bond is formed, the ribosome moves along the mRNA to the next codon.
10. The first tRNA, now at the “E” site, is then released into the cytoplasm for it to be recycled.
11. The second amino acyl-tRNA complex thus moves from the “A” site to the “P” site, leaving the “A” site empty for the next amino acyl-tRNA complex, with anticodon complementary to the third codon along the mRNA.
12. The process is repeated until the ribosome reaches the “stop” codon on the mRNA.

Termination of polypeptide chain

13. Polypeptide chain termination occurs when any of the three stop codons, UAA, UAG or UGA occupies the A site on the ribosome.

14. No tRNAs carry anticodons complementary to the stop codons.
15. Instead, soluble proteins called release factors (RFs) recognise the stop codons and binds to the E site. This brings polypeptide synthesis to a halt.
16. The polypeptide is then released from the ribosome and it assumes its secondary and tertiary structures automatically. The polypeptide may also undergo modification at the Golgi apparatus.
17. The ribosome dissociates from the mRNA, and then disassembles into its subunits.
18. The tRNA molecules are attached again to their respective amino acids.

(c) Discuss how the synthesis of a functional protein can be controlled at the translational and post-translational level. [7]

Regulation at the initiation of translational stage (p. 46 of O&C Euk lect. notes):

1. Availability of eukaryotic initiation factors (eIFs)
 - a. eIFs are additional proteins that promote the proper association of ribosomes to mRNA and are required for protein synthesis to take place.
2. Binding of translational repressors to the 5' cap or 5' untranslated region (UTR) (leader)
 - a. This prevents binding of small ribosomal subunit to 5' cap
3. Binding of translational repressors to the poly(A) tail
 - a. Prevents eukaryotic initiation factors (eIFs) on the poly(A) tail from binding to those on the 5' cap, thus preventing the small ribosomal subunit from binding to the 5' cap

Regulation at the elongation of translational stage:

1. Rate of protein synthesis can also be regulated by the availability of amino acids and regulatory proteins required for translation.

Other points also accepted:

- B1.** The presence of polysomes / polyribosomes can speed up the rate of translation for a particular mRNA.
- B2.** Rate of mRNA degradation will affect the level of translation. If mRNA are degraded, translation will not take place.
- B4.** Rate of mRNA processing will affect the rate of mature mRNA which will affect the rate of translation.

Regulation at post-translational stage : (p. 47 of O&C Euk lect. notes):

4. **Chemical modification** by addition of chemical groups to the polypeptide chain:
 - a. **Phosphorylation** – addition of phosphate group (can activate or deactivate protein, depending on type of protein)
 - b. **Glycosylation** – addition of one or more sugar monomers to form glycoproteins
 - c. **Ubiquitination** – addition of a ubiquitin (a small protein) which identifies the protein for degradation
5. **Modification of the sequence** of amino acids in the polypeptide chain:
 - o Removal of inhibitory portions of the polypeptide chain by protease, for proteins which are synthesized as a larger protein precursor before it can be activated.
6. **Protein degradation**
 - o The longer a protein stays in the cell before it is degraded, the longer it can carry out its functions, thus the expression of the gene that codes for this protein is higher.

[Total: 20]

END OF SECTION B